

Q7

By Work-energy Theorem, work done by the force $W =$ gain in kinetic energy ΔK

Thus $Fx = 6 - 2 = 4$

$$\frac{\Delta K'}{\Delta K} = \frac{W'}{W} = \frac{(2F)(2x)}{Fx} = 4$$

$$\Delta K' = (4)(4) = 16 \text{ J}$$

Hence, final kinetic energy $= 16 + 2 = 18 \text{ J}$

Ans: C

